

Day 6 Assignment — Course Recap Generator

AI Coding Techniques: Build an End-to-End Project Using an Agent Harness

What You're Building

A small tool that takes in PDF slides from this course (your choice of how much — one day's deck, a few days, or a full week) and produces an automated **summary** of that content, along with **diagrams** that visualize it.

You already know this material — you sat through it. That's deliberate. You're the best judge of whether your own tool's output is actually good, because you can tell immediately if it missed something important or got something wrong.

The Harness

Use **Codex** or **OpenCode** — your choice between the two. This is the agent harness that will actually build and run your project with you, the same way we used Claude Code in class.

Required Components

Your project **must** include all four of the following. Everything beyond this list is up to you.

1. A Planning Phase

Before any code gets written, use **plan mode** (if your harness supports it) or **any planning/brainstorming skill** — `obra-superpowers` is one good example, but it's not the only option.

Use this phase to actually decide:

- How will the tool read/extract content from the PDF(s)?
- What should the summary structure look like — per-slide, per-topic, whole-deck?
- What kind of diagram actually fits this content — a concept map, a topic flow, a simple hierarchy?

This isn't a formality. A real plan here will save you real rework later.

2. At Least One MCP Server

Your tool needs to reach outside the harness for at least one real capability — reading the PDF, generating a diagram, or something else you decide fits.

Some good starting points (not mandatory — pick what fits your idea, or find your own):

- A PDF-reading/extraction MCP server

- A diagram-generation MCP server (e.g. something that can turn a structure into an actual rendered diagram, like Mermaid)
- Anything else you find that does real work your project needs

The MCP server should be doing something your project genuinely needs — not bolted on just to check a box.

3. At Least One Harness-Defined Sub-Agent

Your harness lets you declare actual sub-agents — named, with their own role and instructions. You must define and use at least one.

A natural split for this project: one agent summarizes the slide content, a separate sub-agent reviews that summary against the original slides — does it miss a key concept? Misrepresent something? Is the diagram actually faithful to the material?

This needs to be a real sub-agent declared through your harness's sub-agent feature — not just two prompts run one after another.

4. At Least One Custom Skill

Teach your harness a reusable capability — something it can apply consistently, not a one-off instruction.

Example: a skill for "extract key concepts from a slide deck in a consistent format," or "generate diagrams in a consistent style" — something that would still be useful if you fed in a different deck tomorrow.

What's Open to You

- **How much content you feed in** — one day's deck, several, or the whole course so far
- **Which MCP server** — from the suggestions above, or one you find yourself
- **Which planning approach** — plan mode, obra-superpowers, or any other planning skill
- **Any other harness feature** you want to use on top of the four required ones — encouraged, not required. (Think back to Day 6's feature list — agentic editing, codebase context, anything else you found interesting.)

Constraints to Keep in Mind

- **Keep AI credit usage light.** This should be one clean pass — input → MCP action → summarize → review → output — not an open-ended agent loop running indefinitely. A focused, single-purpose tool is the goal, not an elaborate system.
- **Make it something you'd actually want to use again.** A good recap generator for your own notes is a genuinely useful thing to have — and a legitimate project to talk about

later, including on a resume, because you'll actually be able to explain *why* you built it and how it works.

What a Good Submission Looks Like

- A short writeup or README explaining what the tool does and why you made the choices you made during planning
 - Evidence of your planning phase (a plan-mode output, or your brainstorming notes)
 - The MCP server you used and what it does for your project
 - Your sub-agent definition(s) and what each one is responsible for
 - Your custom skill and what it does
 - At least one example run — input PDF(s) in, summary + diagram out
-

Quick Checklist

- ☐ Chose Codex or OpenCode as the harness
- ☐ Used plan mode or a planning skill before building
- ☐ Used at least one MCP server for a real capability
- ☐ Defined and used at least one harness-level sub-agent
- ☐ Created and used at least one custom skill
- ☐ Kept the project to a single clean pass (not an open-ended loop)
- ☐ Ran it end-to-end on at least one real course deck
- ☐ Wrote a short explanation of what you built and why